

ON THE DIOPHANTINE EQUATION $\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$

By

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(Received Dec. 20, 1964)

Introduction. P. Erdős and E. G. Straus conjectured that if $n > 1$ then the Diophantine equation

$$(\star) \quad \frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

has a solution in positive integers x, y, z .

It is known that $(\star)$ has a solution for smaller values of n , for instance, for $1 < n < 8000$, as is seen in [1]. In this connection the present author has announced [3] the existence of solutions of $(\star)$ for $1 < n < 10,000,000$, verified on the electronic computer OKITAC 5090 H of Kyushu University. This paper is a somewhat detailed report on that subject.

It is interesting to note a connection of the Diophantine equation $(\star)$ and the theory of quadratic number fields, in particular, of the Kronecker symbol (§2 below). This connection is peculiar to the numerator 4 at the left-hand side member of $(\star)$, and if we replace this by 5, 6, ..., the connection seems to be less apparent.

§1. Transformation of the problem.

1.—If n is even, then the Diophantine equation $(\star)$ is solvable. More generally we may restrict the problem to the case where $n=p$ is a prime number.

LEMMA 1. *If $n=p$ is a prime, then the Diophantine equation $(\star)$ has a solution if and only if one of the following two congruences has a pairwise prime positive solutions (a, b, c) :*

$$\begin{aligned} a+pb+pc &\equiv 0 \pmod{4abc}, & abc &\not\equiv 0 \pmod{p}; \\ pa+b+c &\equiv 0 \pmod{4abc}, & abc &\not\equiv 0 \pmod{p}. \end{aligned}$$

PROOF. If $n=p$ in (★), exactly one or exactly two of the three integers x, y, z are divisible by p , so that at least one of the two Diophantine equations

$$(1) \quad \frac{4}{p} = \frac{1}{px} + \frac{1}{y} + \frac{1}{z}, \quad xyz \not\equiv 0 \pmod{p};$$

$$(2) \quad \frac{4}{p} = \frac{1}{x} + \frac{1}{py} + \frac{1}{pz} \quad xyz \not\equiv 0 \pmod{p}$$

has a solution.

Let $(y, z)=d, y=db, z=dc$ in (1). Then (1) becomes $dbc+pbx+pcx=4dbcx$, so that $b|x$ and $c|x$. Therefore by putting $x=bcm$ we have $d+pm(b+c)=4dmbc$, so that $d=ma$ for some a with $a+p(b+c)=4mabc$. Conversely the last equation implies (1). Similarly (2) has a solution if and only if $pa+b+c=4mabc$ for some a, b, c and m .

2.—Denote by N_2 the totality of positive odd numbers. If m is a positive integer, $r=\frac{b}{a}$ a reduced rational such that $(m, ab)=1$, we denote by $\left\{ \frac{r}{m} \right\}$ the subset of N_2 , consisting of $x \in N_2$ such that $x \equiv r \pmod{m}$, i.e., $ax \equiv b \pmod{m}$. By using this notation we have

LEMMA 2. *The Diophantine equation (★) for $n=p$, p being an odd prime, has a solution if and only if p belongs either to*

$$(3) \quad \left\{ \frac{-\frac{b}{a}}{q} \right\} \quad \text{with } q+1 \equiv 0 \pmod{4ab}$$

or to

$$(4) \quad \left\{ \frac{-q}{4ab} \right\} \cap \left\{ \frac{-\frac{b}{a}}{q} \right\}$$

for some positive integers a, b, q .

PROOF. If in Lemma 1, $pa+b+c=4mabc$, then by putting $q=4mab-1$ we have $p \in \left\{ \frac{-\frac{b}{a}}{q} \right\}$ with $q+1=4mab$. The condition is also sufficient. If $a+pb+pc=4mabc$, on the other hand, then by putting $q=4mab-p$ we have $p \in \left\{ \frac{-\frac{b}{a}}{q} \right\}$ and $p \in \left\{ \frac{-q}{4ab} \right\}$, and again the condition is sufficient.

Still another formulation in the same line is the following

LEMMA 3. *The Diophantine equation (★) for $n=p$, p being an odd prime, has a solution if and only if p belongs either to*

$$(5) \quad \left\{ \frac{-s}{4ab} \right\} \quad \text{with } s|a+b,$$

or to

$$(6) \quad \left\{ \frac{-\frac{1}{s}}{4ab} \right\} \quad \text{with } s|a+b,$$

for some positive integers a, b, s .

PROOF. Returning to Lemma 1, if $pa+b+c=4mabc$, then $b+c=as$ for some s with $p+s=4mbc$, i.e., $p \in \left\{ \frac{-s}{4bc} \right\}$ with $s|b+c$. The condition is also sufficient. Similarly the condition $a+pb+pc=4mabc$ is equivalent to $p \in \left\{ \frac{-\frac{1}{s}}{4bc} \right\}$ with $s|b+c$.

We summarize the Lemmas above in

THEOREM 1. *The Diophantine equation (★) for $n=p$, p being an odd prime has a solution if and only if p belongs to the union S of the system Σ of coverings $\left\{ \frac{r}{m} \right\}$ of N_2 , where the system Σ consists of*

$$(3) \quad \left\{ \frac{-\frac{b}{a}}{q} \right\} \quad \text{with } q+1 \equiv 0 \pmod{4ab},$$

$$(4) \quad \left\{ \frac{-s}{4abq} \right\} \quad \text{with } s \equiv q \pmod{4ab}, \quad s \equiv \frac{b}{a} \pmod{q},$$

$$(5) \quad \left\{ \frac{-s}{4ab} \right\} \quad \text{with } s|a+b,$$

$$(6) \quad \left\{ \frac{-\frac{1}{s}}{4ab} \right\} \quad \text{with } s|a+b.$$

for some positive integers a, b, q, s .

Actually, the union S does not cover any perfect square (Theorem 2).

§ 2. Connection with Kronecker symbol.

3.—The simplest instance $a=b=q=1$ of (4), and the simplest instance $a=b=s=1$ of (5) and (6) give the same thing $\left\{\frac{-1}{4}\right\}$. Namely if $n \equiv -1 \pmod{4}$, then (★) has a solution. Therefore we restrict ourselves to a study of the set N_4 of all positive integers n satisfying $n \equiv 1 \pmod{4}$, in the rest of this paper. The coverings (3), (4), (5), (6) should be restricted to N_4 , but we retain the symbol $\left\{\frac{r}{m}\right\}$ unchanged, also the system will be denoted by Σ as before, and the union of coverings in Σ by the same symbol S .

In this connection it is to be noted that some coverings may turn to be empty, for example when $q \equiv 1 \pmod{4}$ in (4) or when $s \equiv 1 \pmod{4}$ in (5) or in (6). We exclude these by requiring that $q \equiv -1 \pmod{4}$ in (4), and $s \equiv -1 \pmod{4}$ in (5) and in (6).

4.—Kronecker symbol—Let m be a positive integer, c be an integer $\neq 0$ such that $(m, c)=1$. Then the Kronecker symbol $\left(\frac{c}{m}\right)$ is defined as follows. First if m is odd, then $\left(\frac{c}{m}\right)$ is merely the Jacobi symbol. Next if $m=2^\mu m_0$, with m_0 odd, $\mu \geq 1$, then c is odd and

$$\left(\frac{c}{m}\right) = \left(\frac{2}{c}\right)^\mu \left(\frac{c}{m_0}\right),$$

where $\left(\frac{c}{m_0}\right)$ is the Jacobi symbol and $\left(\frac{2}{c}\right) = (-1)^{\frac{c^2-1}{8}}$,

The Kronecker symbol $\left(\frac{c}{m}\right)$ is a real residue character with conductor m_0 if $m=2^\mu m_0$, m_0 odd, μ even, and with conductor $8m_0$ if μ is odd. For this symbol the following properties are well known.

$$(7) \quad \text{if } c_1 \equiv c_2 \pmod{m} \text{ then } \left(\frac{c_1}{m}\right) = \left(\frac{c_2}{m}\right) \quad \text{for } m \text{ odd or} \\ \text{for } m \text{ a multiple of } 8,$$

$$(8) \quad \left(\frac{c_1 c_2}{m}\right) = \left(\frac{c_1}{m}\right) \left(\frac{c_2}{m}\right),$$

$$(9) \quad \left(\frac{c}{m_1 m_2}\right) = \left(\frac{c}{m_1}\right) \left(\frac{c}{m_2}\right).$$

The reciprocity law of Kronecker symbol may be stated in the form

$$(10) \quad \left(\frac{c}{m}\right) = \left(\frac{m}{c}\right) \quad \text{if } c \text{ is positive and } c \equiv 1 \pmod{4},$$

$$(10') \quad \left(\frac{-c}{m}\right) = \left(\frac{m}{c}\right) \quad \text{if } c \text{ is positive and } c \equiv -1 \pmod{4}.$$

In particular, if m is positive and odd, then $\left(\frac{c}{m}\right)$ was Jacobi symbol, so the symbol may be extended to the rationals $r = \frac{b}{a}$ with $(m, ab) = (a, b) = 1$ by

$$\left(\frac{\frac{b}{a}}{m}\right) = \left(\frac{ab}{m}\right).$$

The same process is possible when m is positive and even, since if a and b are odd then we have $a^2 \equiv 1 \pmod{8}$, or $a \equiv \frac{1}{a} \pmod{8}$.

5.—The following Lemma establishes a close connection of the system Σ of coverings and the Kronecker symbol.

LEMMA 4. For a “significant” (i.e., nonempty) covering $\left\{\frac{r}{m}\right\}$ in the system Σ we have $\left(\frac{r}{m}\right) = -1$.

PROOF. We verify this fact for coverings (3), (4), (5), (6) successively. For (3) we have, by using (10'), that

$$\left(\frac{-\frac{b}{a}}{q}\right) = \left(\frac{-ab}{q}\right) = \left(\frac{-1}{q}\right) \left(\frac{ab}{q}\right) = -\left(\frac{ab}{q}\right) = -\left(\frac{-q}{ab}\right) = -\left(\frac{1}{ab}\right) = -1.$$

For (4), we have to assume $q \equiv -1 \pmod{4}$ since the covering should be nonempty, and we have

$$\left(\frac{s}{4abq}\right) = \left(\frac{-q}{4ab}\right) \left(\frac{-\frac{b}{a}}{q}\right) = \left(\frac{-q}{ab}\right) \left(\frac{-1}{q}\right) \left(\frac{ab}{q}\right) = -1,$$

by using (10'). For (5) we have to assume $s \equiv -1 \pmod{4}$, and by using (10') we have

$$\left(\frac{-s}{4ab}\right) = \left(\frac{-s}{ab}\right) = \left(\frac{ab}{s}\right) = \left(\frac{a}{s}\right) \left(\frac{b}{s}\right) = \left(\frac{a}{s}\right) \left(\frac{-a}{s}\right) = \left(\frac{-1}{s}\right) = -1.$$

The same applies to (6).

THEOREM 2. The union S of the system Σ of coverings does not contain any perfect square.

PROOF. If $n \in S$, then $n \in \left\{ \frac{r}{m} \right\}$ for some $\left\{ \frac{r}{m} \right\} \in \Sigma$ and $\left\{ \frac{r}{m} \right\}$ is not empty. We have seen $\left(\frac{r}{m} \right) = -1$ in Lemma 4. So $\left(\frac{n}{m} \right) = \left(\frac{r}{m} \right) = -1$, and this implies that n cannot be a perfect square.

In [1] it was conjectured that S covers the whole space N_4 , but this is not true by the Theorem 1.

§ 3. Further restrictions.

6.—We take up some simple members of Σ :

$$(12) \quad \left\{ \frac{-1}{3} \right\} \quad (q=3 \text{ in } (3)),$$

$$(13) \quad \left\{ \frac{-1}{7} \right\}, \quad \left\{ \frac{-2}{7} \right\}, \quad \left\{ \frac{-4}{7} \right\} \quad (q=7 \text{ in } (3)),$$

$$(14) \quad \left\{ \frac{5}{8} \right\} \quad (s=3 \text{ in } (5)),$$

$$(15) \quad \left\{ \frac{-2}{15} \right\} = \left\{ \frac{-2}{5} \right\} \cap \left\{ \frac{1}{3} \right\}, \quad \left\{ \frac{-8}{15} \right\} = \left\{ \frac{-3}{5} \right\} \cap \left\{ \frac{1}{3} \right\} \quad (q=15 \text{ in } (3)).$$

Combining these we may say that (★) has a solution unless n is a quadratic residue of 3, 5, 7, and 8.

We have to consider only those n such that n is a quadratic residues of 3, 5, 7 and 8, or such that

$$(16) \quad n \equiv 1, 121, 361, 169, 289, 529 \pmod{840}.$$

By the way the smallest prime with this property is 1009.

Now denote the set of all positive integers n satisfying (16) by N_0 . $n \in N_0$ satisfies $n \equiv 1 \pmod{24}$. If a covering $\left\{ \frac{r}{m} \right\}$ of Σ is such that $m = tm_1$, $(t, m_1) = 1$, $\left\{ \frac{r}{m} \right\} = \left\{ \frac{r}{t} \right\} \cap \left\{ \frac{r}{m_1} \right\}$, and $\left\{ \frac{r}{t} \right\} \supset N_0$, then we can replace $\left\{ \frac{r}{m} \right\}$ by a simpler $\left\{ \frac{r}{m_1} \right\}$. The set of the coverings $\left\{ \frac{r}{m_1} \right\}$ thus obtained will be denoted by Σ_0 . Obviously $\Sigma_0 \supset \Sigma$, since $t=1$ is always permissible.

THEOREM 3. *Let m and q be positive and $(4m, q) = 1$, let $r = \frac{b}{a}$ be a reduced rational such that $(4m, ab) = 1$, and let $m = cd$ with positive integers c, d . If s is positive and $s|cq + d$, $4m|as + b$ then*

$$\left\{ \frac{r^{\pm 1}}{4m} \right\} \cap \left\{ \frac{-s^{\pm 1}}{q} \right\} \in \Sigma.$$

PROOF. By taking cq and d in place of a, b in (5), (6), we have that

$$\Sigma \ni \left\{ \frac{-s^{\pm 1}}{4cdq} \right\} = \left\{ \frac{-s^{\pm 1}}{4m} \right\} \cap \left\{ \frac{-s^{\pm 1}}{q} \right\} = \left\{ \frac{r^{\pm 1}}{4m} \right\} \cap \left\{ \frac{-s^{\pm 1}}{q} \right\}.$$

If in particular $m|b$, and $r=1$ then $\left\{ \frac{-s^{\pm 1}}{q} \right\} \in \Sigma_0$. More precisely:

(i) If q is odd, $s \equiv -1 \pmod{4}$ and $s|q+1$ then $\left\{ \frac{-s^{\pm 1}}{q} \right\} \in \Sigma_0$.

(ii) If q is odd, $s \equiv -1 \pmod{8}$ with $s|q+2$ or $s|2q+1$ then $\left\{ \frac{-s^{\pm 1}}{q} \right\} \in \Sigma_0$.

(iii) If q is odd, $s \equiv -1 \pmod{12}$ with $s|q+3$ or $s|3q+1$ then $\left\{ \frac{-s^{\pm 1}}{q} \right\} \in \Sigma_0$.

(iv) If q is odd, $s \equiv -1 \pmod{24}$ with $s|q+6$, $s|2q+3$, $s|3q+2$ or $s|6q+1$ then $\left\{ \frac{-s^{\pm 1}}{q} \right\} \in \Sigma_0$.

We add here analogous result from (3).

(v) If $q \equiv -1 \pmod{4}$ and $s \mid \frac{q+1}{4}$ then $\left\{ \frac{-s^{\pm 1}}{q} \right\} \in \Sigma_0$.

(vi) If $q \equiv 1 \pmod{4}$ and $s \mid \frac{3q+1}{4}$ with $s \equiv -1 \pmod{3}$, then $\left\{ \frac{-s^{\pm 1}}{q} \right\} \in \Sigma_0$.

We denote the system of coverings obtained by (i)–(vi) by Σ_1 . For this family we see that $\left\{ \frac{q^{\pm 1}}{s} \right\} \in \Sigma_0$ together with $\left\{ \frac{-s^{\pm 1}}{q} \right\} \in \Sigma_0$. This kind of “reciprocity law” does not seem to apply in general. However, we note the following “inverse property”.

THEOREM 4. *If q is a positive odd integer, s_1 and s_2 are positive integers, such that $s_1 s_2 \equiv 1 \pmod{q}$, then $\left\{ \frac{-s_1}{q} \right\} \in \Sigma_1$ if and only if $\left\{ \frac{-s_2}{q} \right\} \in \Sigma_1$.*

If s is a positive integer, q_1 and q_2 are positive odd integers such that $q_1 q_2 \equiv 1 \pmod{4s}$ then $\left\{ \frac{-s}{q_1} \right\} \in \Sigma_1$ if and only if $\left\{ \frac{-s}{q_2} \right\} \in \Sigma_1$.

PROOF. This is checked directly with definition (i)–(vi) of Σ_1 .

7.—We list several simple coverings in Σ_1 for smaller numerators. Note that q should be positive and odd.

$$(16) \quad \left\{ \frac{-1}{q} \right\} \quad \text{for } q \equiv -1 \pmod{4},$$

$$\begin{aligned}
(17) \quad & \left\{ \frac{-4}{q} \right\} && \text{for } q \equiv -1 \pmod{4}, \\
(18) \quad & \left\{ \frac{-2}{q} \right\} && \text{for } q \equiv -1, 5 \pmod{8}, \\
(19) \quad & \left\{ \frac{-3}{q} \right\} && \text{for } q \equiv -1 \pmod{3}, \\
(20) \quad & \left\{ \frac{-7}{q} \right\} && \text{for } q \equiv -1, -2, -4 \pmod{7}, \\
(21) \quad & \left\{ \frac{-8}{q} \right\} && \text{for } q \equiv -1, 5 \pmod{8}
\end{aligned}$$

We may change the numerator from $-s$ to $-s^{-1}$, as is stated in Theorem 4. By the way (17) follows from (i) with $s = \frac{q+1}{4}$, and the second coverings of (19), (21) from (vi) by takings $s=2, 8$ respectively. Note that (16)–(21)

Table 1. Table of $\left\{ \frac{-s}{q} \right\}$ in Σ_1 , q a prime between 11 and 97.

q	s
11	1, 3, 4.
13	2, 5, 7, 8.
17	3, 5, 6, 7.
19	1, 4, 5, 7, 11.
23	1, 2, 3, 4, 6, 8, 12, 13, 16.
29	2, 3, 8, 10, 11, 15.
31	1, 2, 4, 7, 8, 9, 16.
37	2, 5, 8, 14, 15, 19.
41	3, 6, 7, 11, 14, 15.
43	1, 4, 11, 15, 23.
47	1, 2, 3, 4, 6, 7, 8, 12, 16, 17, 24, 25, 27, 32, 36.
53	2, 3, 5, 8, 18, 20, 23, 27, 30, 32.
59	1, 3, 4, 5, 7, 12, 15, 17, 20, 36, 41.
61	2, 8, 23, 31.
67	1, 4, 9, 15, 17, 23, 35.
71	1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 36, 37, 40, 48.
73	2, 5, 11, 15, 20, 21, 39, 44.
79	1, 2, 4, 5, 8, 10, 16, 20, 21, 23, 32, 40, 42, 55, 64.
83	1, 3, 4, 7, 12, 21, 28, 30, 36.
89	3, 6, 7, 15, 23, 30, 31, 51.
97	5, 7, 13, 14, 15, 39.

Table 2. Table of $\left\{\frac{s-1}{q}\right\}$ in $\sum_1$ for $s=1, 2, 4, 8$.

s	q
1, 4	$q \equiv -1 \pmod{4}$.
2, 8	$q \equiv 5 \text{ or } 7 \pmod{8}$.

 Table 3. Table of $\left\{\frac{-s+1}{q}\right\}$ in $\sum_1$ for $s \equiv -1 \pmod{4}$, q odd.

s	q
3	$q \equiv -1 \pmod{3}$
7	$q \equiv -1, -2, -4 \pmod{7}$
11	$q \equiv -1, -3, -4 \pmod{11}$
19	$q \equiv -1 \pmod{19}$
23	$q \equiv -1, -2, -3, -4, -6, -8, -12, -13, -16 \pmod{23}$
31	$q \equiv -1, -2, -16 \pmod{31}$
43	$q \equiv -1 \pmod{43}$
47	$q \equiv -1, -2, -3, -6, -8, -16, -24, -25, -32 \pmod{47}$
59	$q \equiv -1, -3, -20 \pmod{59}$
67	$q \equiv -1 \pmod{67}$
71	$q \equiv -1, -2, -3, -6, -12, -24, -36, -37, -48 \pmod{71}$
79	$q \equiv -1, -2, -40 \pmod{79}$
83	$q \equiv -1, -3, -28 \pmod{83}$

show that all the "coverings" $\left\{\frac{-s}{q}\right\}$ with the Jacobi symbol $\left(\frac{-s}{q}\right) = -1$ are contained in $\sum_1$, for $s=1, 2, 3, 4, 7$ and 8 .

We attach Tables 1 and 2, 3 to show the system $\sum_1$, when classified by the denominator q and by the numerator s . In Table 1 only prime denominators less than 100 are given.

§ 4. Computations.

8. We verified the solvability of (★) for $1 < n < 10^7$ by the electronic computer OKITAC 5090H of Kyushu University. The programming responsibility goes to the author. The procedure was as follows.

1) Only the quadratic residues of 3, 5, 7, and 8 are treated. Thus n satisfies

$$1 < n < 10^7,$$

$$n \equiv 1, 121, 361, 169, 289 \text{ or } 529 \pmod{840}.$$

2) Perfect squares are discarded.

3) If n is either divisible by 11, 13, 19, 23 or 47, or congruent to some number found in the corresponding rows of Table 1, then n is discarded.

4) The procedure corresponding to the "complementary law type" coverings.

a) If n , $n+1$, $n+4$ or $4n+1$ is divisible by some integer m such that

$$\begin{aligned} 1 < m < 10^3, \\ m &\equiv 3 \pmod{4} \end{aligned}$$

then n was printed out with m .

b) If n , $n+2$, $n+8$, $2n+1$ or $8n+1$ is divisible by some integer m such that

$$\begin{aligned} 1 < m < 10^3, \\ m &\equiv 5 \text{ or } \equiv 7 \pmod{8}, \end{aligned}$$

then n was printed out with m .

5) The procedure corresponding to the "reciprocity law type" coverings.

For $s=3, 7, 11, 19, 23$ and 47 the computer was to determine the remainder of n when divided by integers m found in the corresponding rows of Table 3, such that

$$1 < m < 10^3.$$

If the remainder is either $=m-s$ or congruent to $-\frac{1}{s} \pmod{m}$, then n was printed out with s and m .

The computing time was 122 minutes and the result was that only two numbers survived through all this process. These are 1095481 and 7502881, both being primes incidentally. Now 1095481 belongs to the covering $\left\{-\frac{59}{589}\right\}$, $589 \equiv -1 \pmod{59}$ of the "reciprocity law type", and 7502881 belongs to the covering $\left\{-\frac{8}{1117}\right\}$, $1117 \equiv 5 \pmod{8}$ of the "complementary law type". Therefore the established fact is that *if n is a quadratic residue of 3, 5, 7, 8 such that $n < 10^7$ and if n is not a perfect square, then either n has a divisor $d \neq n$ such that $1 < d < 10^3$, or n belongs to some covering $\left\{\frac{r}{m}\right\}$ in the*

system Σ_1 . In particular, the solvability of the Diophantine equation (★) was verified for all n in the range $1 < n < 10^7$.

Note that the primality of n was not so essential in our computation. So it is believed that any positive integer, which is a quadratic residue of 3, 5, 7 and 8, and still is not a perfect square, is contained in some covering of the system Σ_1 .

References.

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